Superstore Sales Analytics Report

1. Introduction

In this project I built an interactive sales dashboard in Power BI using the Superstore dataset. My goals were to practise the full analytics workflow (preparing data, building a data model, writing DAX measures and creating charts) and to find out how sales and profit changed from 2020 to 2023. I then checked the results myself and explained them in this report.

2. Dataset and tools

The data covers orders from January 2020 to December 2023.

Item	Detail
Dataset	Superstore sales data [Kaggle]
Period	3 Jan 2020 to 30 Dec 2023
Size	10,194 rows in the Orders table
Tables used	Orders, Returns, Date table, key measures
Excel	Looking at the data and cleaning it
Power BI Desktop	Data model, DAX measures and dashboard

3. Data model

A data model tells Power BI how the tables are connected. Mine has four tables:

- **Orders:** the main table with every sale.
- **Date table:** a calendar linked to Orders by Order Date. This is what lets me compare sales with the previous year.
- **Returns:** the list of returned Order IDs, linked to Orders by Order ID. It is used to work out the percentage of returned orders.
- **key measures:** a table that only stores my calculations (DAX measures).

4. Key metrics

The dashboard compares each number with the previous year (PY). Sales and profit both grew by almost half.

Metric	Current	Previous year	Change
Sales	$2.33M	$1.58M	+47.16%
Profit	$292.30K	$196.37K	+48.85%
Returned orders	5.79%	8.74%	-2.95%

5. Dashboard overview

The dashboard has one page. The top row has three KPI cards, and below them are four charts. The slicers on the left let a user filter everything by customer, country, segment and date.

- **KPI cards:** sales, profit and percentage of returned orders, with the previous-year value and the change.
- **Profit by Product:** profit of each sub-category. Red bars are the lowest profit and green bars are the highest.
- **Profit by state:** a US map where darker green means higher profit.
- **Sales vs Previous year over time:** monthly sales compared with the same month a year earlier.
- **Sales by Segment:** how sales are split between customer segments.

6. Key insights

1. **Sales and profit grew strongly.** Sales went from $1.58M to $2.33M (+47.16%), and profit went from $196.37K to $292.30K (+48.85%). Profit grew a little faster than sales.
2. **Sales trend upward.** Monthly sales go up from 2020 to 2023, with the highest peak in late 2023 at above $100K for one month.
3. **Some states matter more.** California and New York are the most profitable states on the map. Texas is one of the least profitable.

4. **One segment brings half of sales.** The segments account for (Consumer) 50.32%, (Corporate) 30.77% and (Home office) 18.92% of sales.
5. **Product profit is uneven.** Binders, Paper, Phones and Accessories have high profit, while Tables, Bookcases and Supplies have the lowest [check each by hovering over the bars].
6. **Returns are low but worth watching.** 5.79% of orders were returned, which is 2.95% lower than the previous year.

7. Recommendations

These suggestions come from the insights above.

1. **Check low-profit products.** Look at the prices and discounts of products such as Tables before running more promotions.
2. **Focus on strong areas.** Put more marketing effort into the biggest customer segment and the most profitable states, California and New York.
3. **Study weak areas.** Find out why Texas is less profitable, for example because of discounts or product mix.
4. **Reduce returns.** Look at which orders are returned and why, to bring the 5.79% return rate down.
5. **Prepare for busy periods.** Plan stock and staff for the sales peaks near the end of each year.

8. Limitations

- This is a one-page dashboard built on one dataset, so it does not cover every question about the business.
- Profit is used as given. The data has no separate cost or marketing figures.
- The dashboard shows what happened, not why. Reasons, such as why Texas is weak, need more analysis.